



Ranking on user–item heterogeneous graph for Ecommerce next basket recommendations

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ARTICLE INFO

Keywords:

E-commerce
Recommender systems
Sequential recommendation
Graph ranking
Random walk

ABSTRACT

Recommender systems have become an indispensable engine for online applications, which can help users locate the items they need among numerous other candidate items. Despite the success of general models, sequential recommender systems based on sequential patterns have garnered attention, especially in e-commerce, because customers do not necessarily purchase the same items over time. Markov chains and neural networks are the most adopted models in previous studies, but few integrated user sequential order information and user numeric rating feedback. To simultaneously capture the time-aware demands and time-invariant interests of e-commerce customers, this study proposes a unique heterogeneous graph structure to model sequential buying order and rating feedback and develops a query-based recommendation approach to predict future purchase behavior. Experiments on four different datasets show that the approach demonstrates improved performance for different metrics.

1. Introduction

Cross-selling has challenged the e-commerce industry because of the extremely large volume of e-commerce products and users [1]. For instance, Amazon sells hundreds of millions of products to customers worldwide, and China's largest online shopping platform, Taobao, has over 700 million monthly active users. The problem of information overload has led to difficulties among e-commerce customers in locating the right products efficiently. From the perspective of sellers or platforms, general coupons are unable to satisfy the dynamic and personalized demands of customers. Recommender systems have been featured as personalized shopping assistants in large-scale e-commerce sites [2,3], which can increase sales and loyalty and reduce customer churn [1,4–7].

Conventional recommender systems commonly address the problem of “which items will best fit a user's overall interests”, that is, to predict possible favorite movies, groceries, or restaurants for an individual. Such user–item matching tasks have attracted increasing attention and yielded remarkable results in the field of data mining and machine learning [8,9]. However, in e-commerce cross-selling, such tasks can create the problem of “which items are a user most likely to buy in the next time”, resulting in a new sequential recommendation direction [10–12]. Instead of resorting only to users' overall time-invariant interests, sequential recommender systems can also consider potential buying dependency relations between items to predict users'

time-changing demands. Differing from general sequential recommendations, next-basket recommendation models address the sequential order data between item sets, instead of individual items [13,14], and are believed to be highly appropriate in e-commerce, because people may want to purchase more than one item in the same transaction or within a short period of time.

We follow the general assumption of existing studies that user dynamic demands are affected by two factors: users' overall preference for items and the sequential dependency relationship between items. However, one of the shortcomings of related studies is their lack of integration of users' rating feedback. In previous studies, time-aware sequential demands and time-invariant overall preferences were learned from users' binary buying records. Although some models do not consider the rating information because of the nature of the problems they handle (e.g., no feedback or only binary feedback is available in some applications), scaled ratings are believed to represent users' preference for items more precisely than binary buying records. Furthermore, rating feedback has become an increasingly common feature in e-commerce, as shown in Fig. 1. Hence, developing a new e-commerce recommendation model that can handle users' buying order information and rating feedback is essential. In this study, we propose a user–item relational graph and a query-based ranking algorithm for e-commerce sequential recommendations.

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<https://doi.org/10.1016/j.knosys.2024.111863>

Received 1 January 2023; Received in revised form 21 April 2024; Accepted 25 April 2024

Available online 29 April 2024

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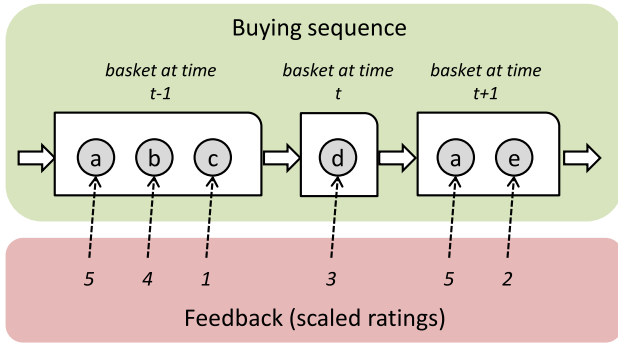


Fig. 1. An example of user sequential buying behavior and feedback in e-commerce.

As illustrated in Fig. 1, our study aims to construct a heterogeneous graph with two types of nodes (users and items) and two types of edges (user-item rating relations and item-item buying dependency relations), and treat the personalized recommendation problem as a ranking problem on vertices of the unique heterogeneous graph. In detail, user-item preference relation is retrieved from users' rating data and item-item dependency relation is retrieved from users' sequential buying data, then the purchase demand of each user at a given time will be modeled as a personalized input query, which represents the user's historical buying and rating behaviors. Our study makes three contributions to the literature. (1) Our study is the first to fully utilize users' scaled rating feedback for e-commerce sequential recommendations. (2) The proposed model is able to naturally capture user buying sequences of multiple items and is adaptable for systems with or without rating functionality. (3) Our study presents a novel iterative learning algorithm for heterogeneous graphs with different node and edge types.

The rest of this paper is structured as follows: Section 2 reviews the works most relevant to this study, including general recommendation models and sequential recommendation models; Section 3 describes the next-basket recommendation problem and how a heterogeneous graph can be created with users' sequential behavior and rating data; Section 4 presents the iterative ranking method for heterogeneous graphs with two types of nodes and edges and discusses how to create a personalized user query for sequential demands; Section 5 reports the results of the comparative study with four different datasets; and Section 6 summarizes the findings and suggests future research avenues.

2. Related works

Recommender systems are a hot topic in the literature. Historical rating data provided by users for items are commonly used for profiling users, which is known as collaborative filtering (CF) [15,16]. Considerable research has been conducted on CF with various techniques to mine precise and deep information for rating predictions or item rankings, such as KNN models [17,18], matrix factorization [19,20], graph ranking [21–24], probabilistic topic model, deep neural networks [25–28], and so on. In recent years, research on sequential recommender systems, which are an emerging type of recommender systems, has increased. General recommender systems recommend items based on users' overall tastes but are incapable of responding to their dynamic demands. However, sequential recommender systems specify users' sequential behavior to make dynamic item suggestions.

The most commonly used models in current research are Markov chains, which assume that users' future decisions are influenced by their most recent action (one-order models) or several actions (multiple-order models). Shani et al. [29] proposed a Markov decision process model and a multigram Markov chain predictor for sequential recommendations. In their model, which can predict future decisions

based entirely on the order in which products are purchased, users' general preferences are ignored. Rendel et al. [13] also created a recommender for sequential data based on personalized Markov chains. By combining Markov chains and matrix factorization, the recommender creates a cube representing each user's individual transition matrix and uses high-order matrix factorization to mine the personalized sequential patterns of each user. Meanwhile, Wang et al. [30] proposed a general aggregation framework for balancing sequential and general preferences. Similar works were presented in [31,32], which developed various Markov chain models to improve recommendation accuracy. Chen et al. [33] employ Residual connections and Graph Neural Networks based on Markov Chain to propose MC-RGN, which incorporates group preferences into a single user-item historical sequence. Some heuristic methods for temporal user behavior prediction also utilize users' implicit auxiliary behavior, such as "clicking stream" [34,35].

Neural networks have recently been introduced to address sequential recommendation problems [10]. Yu et al. [36] presented a dynamic recurrent neural network (RNN) that considers users' current (local) interests and global sequential features to simulate their sequential behavior. In [37], an RNN was also employed to address a unique multi-behavioral sequential prediction task. Furthermore, Tang et al. [38] proposed a convolutional sequence embedding-based recommendation model that involves embedding a short sequence of previously purchased items into an "image", then learning sequential patterns to predict future items. Attention-based networks are also used in sequential recommendations [39,40] and considered to be capable of learning users' short- and long-term interests. Zhang et al. [41] proposed a Time-Aware Transformer for Sequential Recommendation (TAT4SRec), an self-attention based neural network model which utilizes the temporal information to captures users' preferences more precisely.

According to the literature, users' dynamic demands and general tastes are essential for sequential recommendations. However, previous research, such as that on Markov chains and neural network models, considered only user interaction sequences with items and rarely used ratings, which are a common type of feedback in real-world applications. For example, in [38,40], ratings with different values were converted into binary feedback, which resulted in less accurate user profiling, to a certain extent. The primary distinction between our work and prior studies is that we offer a heterogeneous graph model with diverse user-item relationships, which will allow us to incorporate users' purchase sequences and rating feedback information. Heterogeneous graphs have been emerging as useful tools to model complex relationships between different information entities. For example, authors of [42,43] built various relations between students and their exercises as input of a bi-Graph contrastive learning algorithm for the knowledge tracing problem. Our work is similar to these studies in exploring possible entities and relations in the scenario of recommender systems to enhance recommendation accuracy and efficiency.

3. Problem definition and heterogeneous graph construction

3.1. Next-basket recommendation problem

In the case of next-basket recommendations, the products purchased in the last or earlier transactions are used to forecast the user's next transaction. Moreover, the absolute timestamps of customers' purchasing actions are discarded, leaving only the sequence order of the purchased items [13]. Customers typically purchase multiple items at once; thus, specifying the order of individual items in the same transactions would be difficult [38]. Set-level sequential patterns are common in e-commerce, and the related recommendation problem is referred to as next-basket recommendations [13]. A "basket" refers to a group of items purchased in a single transaction by a user. When data are collected, transactions are separated based on the precision level of the timestamps. In this study, we specify the item baskets within an acceptable time period, such as one day or one hour, based on the

Table 1

Notations.

Notations	Meanings
U/I	The user/item set
$B/B^u/B^u$	The basket sequence data of all users/user u /the r th basket of user u
$r(u, i)/r(u, i, t)$	The rating feedback of user u for item i in the last/ t th basket
$a(i, j)$	The sequential transition relation from items i to j
R/A	The edge set of user–item rating/item–item transition relations
$d_r(u)/d_r(i)$	The degree of user u /item i regarding the rating edges R
$d_a^{out}(i)/d_a^{in}(i)$	The out/in degree of item i regarding the transition edges A
S_R/S_A	Regularization matrices of the edge set R/A
$\mathbf{y}/\mathbf{y}(i)$	The query vector/score of item i for target user
$\mathbf{f}/\mathbf{f}(i)$	The predicted ranking vector/score of item i for target user

item domain. Details on the data preprocessing for different datasets are provided in Section 5. Unlike previous studies, our study identifies users' buying behavior as a two-step process of purchasing and rating; hence, we define the next-basket recommendation problem as follows:

Let $U = \{u_1, u_2, \dots\}$ and $I = \{i_1, i_2, \dots\}$ be the user set and item set, respectively. For a particular user u , the purchase history is denoted as $B^u = (B_1^u, B_2^u, \dots)$, which represents the sequence of their buying baskets, where each basket $B_t^u \subset I$ consists of multiple items. The buying sequence data of all the users are denoted as $B = \{B^u\}$, and a rating function $r(u, i, t \in \mathbb{R})$ is associated, denoting the feedback of a given user u for item i in its t th basket. We define the next-basket recommendation problem in this study to seek an optimal ranking function $f(i|u, B, r) \in \mathbb{R}$ that can predict the ranking score of unknown item i of a given target user u . The key notations are in Table 1. Similar to typical recommender systems, the top N items with the highest ranking scores will be chosen as suggestions for the user's next purchase decision. Recommender systems try to predict unknown items; therefore, the proposed next-basket recommendation model will not suggest previously purchased items.

3.2. Construct a heterogeneous graph from buying sequence and rating feedback

We design a heterogeneous graph model using the original sequential data, as well as rating information, to build two types of links between users and items.

3.2.1. Undirected user–item rating relations

A user may provide a rating for the products in each buying basket. If the same item appears in many baskets, then it may receive duplicate ratings. We use the last rating assigned to an item to represent the user–item rating relation, which we formalize as $r(u, i) \in [0, 1]$, with the necessary normalization, and treat as an undirected relationship between a user and an item.

3.2.2. Directed item–item transition relations

The transition probability between items can be retrieved from the user purchase sequences, similar to one-order or higher-order Markov chain models. We present a novel method for establishing a transition network between items to integrate the rating feedback. First, we define the directed relation a between two items i and j , as follows:

$$a(i, j) = \frac{\sum_{l=1}^L \sum_{t: i \in B_{t-1} \& j \in B_t} \frac{r(i, j, t)}{l}}{\max a(i, \cdot)} \in [0, 1] \quad (1)$$

In (1), to control the search depth, we consider the maximally L -level influence on the user sequential decisions. However, such influence is believed to weaken gradually over time, so a distance variable l is introduced as a decay factor. Here the user information is omitted because the item–item transition relation is counted from the sequential information over all users.

Eventually, we construct a directed relational network for all the items, thereby indicating the strength of their purchasing dependency relationships.

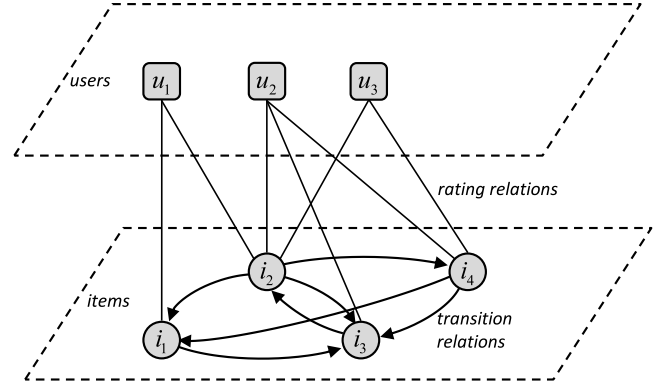


Fig. 2. Example of the heterogeneous graph structure between users and items based on buying sequence and rating feedback.

3.2.3. Graph structure

The combination of the above established user–item rating relation and item–item transition relation yields a heterogeneous graph for users and items. Let $G = (U \cup I, R \cup A)$ be a heterogeneous graph, with users U and items I as the vertex sets and two types of relations, as the edge sets. The first type of edge R , represents the undirected rating relationship between the user and item nodes, whereas the second type of edge A , represents the directed transition relation between the item nodes. The weighting functions of the two edge types are denoted as $R_{ij} = r(u_i, i_j)$ and $A_{ij} = a(i_i, i_j)$, respectively. We normalize all the weightings within the range of $[0, 1]$. Fig. 2 illustrates the heterogeneous graph structure, where we can find that each rating relation connected a pair of user and item, representing the rating information; and each (directed) transition relation connected a pair of items, representing the sequential buying dependency between items.

In addition, for the undirected rating edge R , we define the degree metric as follows:

$$d_r(u) = \sum_{i \in I} r_{ui}, d_r(i) = \sum_{u \in U} r_{ui}. \quad (2)$$

For the directed transition edge A , we define the out-degree and in-degree metrics as:

$$d_a^{out}(i) = \sum_{j \in I} a_{ij}, d_a^{in}(i) = \sum_{j \in I} a_{ji}. \quad (3)$$

4. SeqRank model

The proposed SeqRank model learns a ranking function $f : I \rightarrow \mathbb{R}$ from the proposed heterogeneous graph $G = (U \cup I, R \cup A)$, which maps each item node to a real number based on a request user as input query. The top N items are chosen as the recommendations for the request user. We propose an iterative algorithm as below.

4.1. Ranking manner

An optimal ranking for relational data should adhere to two norms, according to query-based graph ranking theories [44,45]: It is smooth for vertices with strong relationships firstly, and consistent with prior preferences secondly (query). We also offer a vector form $\mathbf{f}$ that represents the ranking score distribution, namely, $\mathbf{f}_i = f(i)$. To ensure smoothness, the ranking of items with strong relationships should not vary considerably. Motivated by prior regularized graph ranking schemes [44], we assume that a particular item node's ranking score is refined as the sum of the contributions from its linked vertices, including both user nodes and item nodes, provided as:

$$f(i) = \mu \sum_{j \neq i} \frac{a_{ji}}{\sqrt{d_a^{\text{out}}(j)d_a^{\text{in}}(i)}} f(j) + (1 - \mu) \sum_{j \neq i} \sum_{u \in U} \frac{r_{ui}}{\sqrt{d_r(u)d_r(i)}} \frac{r_{uj}}{\sqrt{d_r(u)d_r(j)}} f(j). \quad (4)$$

In (4), we import the regularization framework of [44] to “smooth an edge weight by the degree of its two connected vertices simultaneously”. With the above settings, we weight an item based on credits from two types of paths:

1. partly (μ) transited from other items, following the sequential transition relations;
2. partly $(1 - \mu)$ transited from intermediate user nodes, following the rating relations.

The iteration can be seen as a weighted integration of the two input information resources: users' general (personalized) preferences and behavioral sequence information. We provide a random walk model in Section 4.4 for an extensive explanation.

We also define two symmetric regularization matrices based on (4), as follows:

$$S_R = \left[\frac{r_{ui}}{\sqrt{d_r(u)d_r(i)}} \right]_{|U| \times |I|}, \quad S_A = \left[\frac{a_{ij}}{\sqrt{d_a^{\text{out}}(i)d_a^{\text{in}}(j)}} \right]_{|I| \times |I|}. \quad (5)$$

Subsequently, a personalized query vector $\mathbf{y}$ is required for the target user as the prior knowledge input. Section 4.2 describes how a personalized query vector is initialized.

Based on the above notations, the iterative updating rule for the entire ranking vector is presented as follows:

$$\mathbf{f} = \alpha (\mu S_A^T + (1 - \mu) S_R^T S_R) \mathbf{f} + (1 - \alpha) \mathbf{y}. \quad (6)$$

Here, parameter $\alpha \in (0, 1)$ adjusts the contribution of the graph structure versus that of the prior query input, similar to the random walk with restarts theory; whereas the parameter $\mu \in (0, 1)$ adjusts the contribution of the sequential effects versus that of the users' general interests, referring to (4).

The ranking distribution $\mathbf{f}$ can be random in the beginning, and we employ (6) to refine it repeatedly. The convergence is well discussed and proven in similar models, such as a random walk with restart [46], the regularization framework of [44] and the BiRank model established by He et al. [21].

4.2. Personalized query

The SeqRank model also requires the query vector $\mathbf{y}$, which represents users' personalized demands, as prior knowledge. We initialize the query vector, following two basic criteria, to effectively integrate static and dynamic user demands:

Rule 1: More recently purchased items are preferred;

Rule 2: Items with higher ratings are preferred.

The first rule emphasizes users' most recent choices, whereas the second rule considers users' general preference for an item. In detail, for target user u with historical buying sequences in terms of baskets

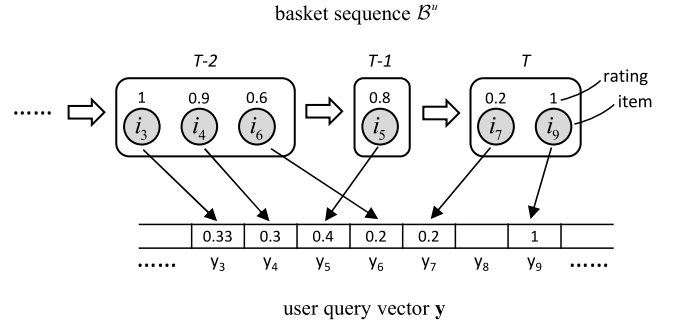


Fig. 3. Initialization for the personalized query vector of a user.

$B^u = (B_1^u, B_2^u, \dots)$, we set the score of item i in the user's query vector $\mathbf{y}$ to $y_i = r(u, i)/\delta_i$ if $i \in B^u$ and 0 otherwise. The offset factor δ_i denotes the distance of the last basket with item i to the overall final basket B_T^u .

Fig. 3 shows an example of query vector construction using the successive buying behavior of a particular user. The query vector need not be normalized to a stochastic vector, because the absolute values have no effect on the ranking outcomes.

4.3. Iterative algorithm

Algorithm 1 provides the details of the SeqRank model for sequential recommendations, where user-item graph G and a query vector $\mathbf{y}$ must be prepared as the given input. Recommender systems aim to discover new items; thus, the top-ranked unpurchased items, according to the ranking results over all items, will be selected as recommendations for the target user's next buying decision.

Input: Heterogeneous graph $G = (U \cup I, R \cup A)$ for all users and items; query vector $\mathbf{y}$ for the target user; parameters α and μ

Output: A ranking vector $\mathbf{f}$ for items

- 1 Construct two symmetrical matrices S_A and S_R from G :

$$S_R = \left[\frac{r_{ui}}{\sqrt{d_r(u)d_r(i)}} \right], \quad S_A = \left[\frac{a_{ij}}{\sqrt{d_a^{\text{out}}(i)d_a^{\text{in}}(j)}} \right]$$
- 2 Randomly initialize $\mathbf{f}$;
- 3 **while** stopping criteria is not met **do**
- 4 $\mathbf{f} \leftarrow \alpha (\mu S_A^T + (1 - \mu) S_R^T S_R) \mathbf{f} + (1 - \alpha) \mathbf{y}$;
- 5 **end**
- 6 **return** $\mathbf{f}$;

Algorithm 1: Algorithm for the SeqRank model

4.4. A random-walk version for extensive explanation

In the SeqRank model, we use the α parameter to balance the effects of the graph structure versus those of the user query and μ to balance the contributions of the sequential effects versus those of the static preferences. A random walk model is presented below for clarity.

Referring to the structure in Fig. 2, given the heterogeneous graph G with two groups of nodes U and I and two types of edges R and A , we implement random walk on the heterogeneous graph structure, starting with the target user node, using the following unique strategy:

- (1) If the currently visited node is a user node, then an adjacent item node will be selected in the next step, following the user-item rating relation.
- (2) If the currently visited node is an item node, then it will again have two options for the next move:
 - (a) Option one: with probability μ that an adjacent item node will be selected, following the item-item transition relation;

- (b) Option two: with probability $(1 - \mu)$ that an adjacent user node will be selected, following the user–item rating relation;

We summarize the conditional probability moving between the user and item nodes as follows: user to item $p(i|u) = r_{ui}/d_r(u)$, item to user $p(u|i) = r_{ui}/d_r(i)$, and item to item $p(j|i) = a_{ij}/d_a^{out}(i)$, where $u \in U$, and $i, j \in I$.

With the above random walk strategy, we present the probability of visiting a user or an item node in the next step as:

$$\begin{cases} p^{(t+1)}(u) &= (1 - \mu) \sum_{i \in I} p(u|i)p^{(t)}(i) \\ p^{(t+1)}(i) &= \sum_{u \in U} p(i|u)p^{(t)}(u) + \mu \sum_{j \in I} p(i|j)p^{(t)}(j) \end{cases} \quad (7)$$

We treat users as intermediate nodes, because recommendation tasks are concerned only about item rankings; thus, we obtain the combined update for only the items, as follows:

$$p^{(t+1)}(i) = \mu \sum_{j \in I} p(i|j)p^{(t)}(j) + (1 - \mu) \sum_{u \in U} \sum_{j \in I} p(i|u)p(u|j)p^{(t)}(j) \quad (8)$$

Let $\mathbf{p}^{(t)}$ denote the visiting distribution over all the items in time t . We obtain the following update according to random walk with restart theory:

$$\mathbf{p}^{(t+1)} = \alpha \left[\mu (D_A^{-1} A)^T + (1 - \mu) (D_R^{-1} R)^T (D_R^{-1} R) \right] \mathbf{p}^{(t)} + (1 - \alpha) \mathbf{p}^{(0)} \quad (9)$$

By comparing (6) and (9), we can see that the SeqRank model is similar to the random walk model but only when the transition matrices $D_R^{-1} R$ and $D_A^{-1} A$ are replaced with the symmetric versions S_R and S_A . Thus, the SeqRank model can be considered as a regularized version of the random walk model. The literature showed that regularized models can outperform random walk models when handling personalized graph ranking tasks [22,31], and our experimental analysis (Section 5) supports this finding.

5. Experiments

We conduct experiments on four public datasets for a comparative analysis of the performance of the SeqRank model in providing next-basket recommendations.

5.1. Datasets and experiment setup

The datasets are from three domains: Movielens.com, Tmall.com, and Amazon.com. Movielens provides various datasets for examining recommender systems, and we use the 1M version in this study, which contains around 1 million rating records. Tmall is one of China's leading B2C shopping sites, and we choose the dataset from the IJCAI 2015 competition,¹ because it contains actual buying records within the span of several months and thus is frequently used for testing sequential recommendation models [30,38,47]. The same, Amazon records users' reviews of products on the site, which are widely used for recommender system evaluation [47]. We select two subcategories from Amazon.com: "Beauty" and "Movies_and_TV" (MoviesTV) from the collection of [48].

Recalling that the next-basket recommendation models handle sequence information between item sets (baskets), rather than individual items, we round off the original timestamp values to appropriate values for each dataset to create buying baskets with reasonable sizes. In detail, the timestamps in the Movielens dataset are rounded off to the nearest 100 s, because Movielens is a pure movie review site that users do not actually buy items on it, as a result, user rating behaviors are of high frequency. For the other three datasets, Tmall, Beauty, and

Table 2

Dataset statistics.

	MovieLens	Tmall	Beauty	MoviesTV
#Users	5789	9780	814	18 060
#Items	3416	13,665	788	33 927
#Ratings	781,056	140,234	16 675	643 521
#Ratings per user	134.9	14.3	20.49	35.6
Sparseness	96.05%	99.89%	97.40%	99.89%
Split threshold	100 s	10 days	10 days	10 days
#Baskets	149,446	72,826	9696	291 523
#Baskets per user	25.8	7.4	11.9	16.1
Avg. basket size	5.2	1.9	1.7	2.2

Table 3

Model type and sources.

Model	Model type	Sequence type	Buying order	Rating feedback
CF	Shallow	Non-sequential	No	Yes
MC	Shallow	Item-level	Yes	No
FPMC	Shallow	Basket-level	Yes	No
Caser	Deep	Item-level	Yes	No
FMLP-Rec	Deep	Item-level	Yes	No
M ²	Deep	Basket-level	Yes	No
SeqRank	Shallow	Basket-level	Yes	Yes

MoviesTV, timestamps are rounded off to 10 days. That is, we assume that the products purchased within 10 days are part of the same buying basket.

For each dataset, we keep only the users with at least 15 baskets and the items purchased/selected by at least 5 users for our experiments. Further, each dataset is split according to the sequences of buying baskets of each user: the earliest 80% baskets are for model training and the rest 20% baskets are for testing. Note that only those items with positive ratings (4 and 5 stars in Movielens and Amazon datasets) are remained for testing, for the purpose of testing the ability of each model to predict the actually loved items in the future. Tmall dataset do not have ratings, so all items in the latest 20% basket are selected.

Table 2 records the statistical information of each dataset for model training, which shows that the Movielens is the most dense dataset, with the users having more and larger baskets. Real e-commerce datasets (Tmall, Beauty, MoivesTV), however, have less baskets and ratings.

5.2. Baseline models

We select the following baseline models with distinct algorithms for the comparative study: (1) **CF** – The conventional user-based CF model uses only rating data and thus yields individualized but non-sequential results. (2) **MC** – The simple one-order Markov chain model predicts future items based only on the last buying basket of a user. Thus, it is a sequential but unpersonalized model. (3) **FPMC** – The Factorized Personalized Markov Chain (FPMC) model [13] is selected. It incorporates the effects of both sequential patterns and general user interests. (4) **Caser** – A deep learning model, namely, convolutional sequence embedding recommendation model [38], which constructs a unique convolutional neural network to capture static and dynamic user behavioral patterns. (5) **FMLP-Rec** – One of the state-of-the-art deep model that introduces a filter-enhanced MLP for sequential recommendations [49]. (6) **M2** – One of the latest deep model that incorporates user preferences and global popularities and transitions among items [47]. It has been reported to be effective at capturing user repeated behaviors. (7) **SeqRank-walk** – A variant of SeqRank based on the random walk strategy mentioned in Section 4.4. We test this to evaluate the effectiveness of employing the regularization framework in SeqRank. (8) **SeqRank-norate** – A variant of SeqRank that utilizes no rating information, for the purpose to evaluate the effectiveness of SeqRank in handling scaled rating feedback.

¹ International Joint Conference on Artificial Intelligence (IJCAI2015): Repeat Buyers Prediction Competition, <https://ijcai-15.org/index.php/repeat-buyers-prediction-competition>.

Table 3 summarizes the main differences of compared models. We can find that both shallow and deep models, and both item-level and basket level sequential models are compared, however, rating feedback is omitted in all sequential models except the proposed SeqRank model. Each model has several parameters, and we utilize a grid search to optimize the results on each dataset.

5.3. Evaluation metrics

We select the following metrics from different aspects to assess the performance of the compared models:

Classification accuracy. As usual, we define the *precision* metric for a test user as the proportion of the correctly predicted items in the prediction set, that is, top- n , and the *recall* metric as the proportion of the correctly predicted items in the truly selected items in the test set.

$$prec@n = \frac{|prediction \cap test|}{n}, \quad recall@n = \frac{|prediction \cap test|}{|test|}$$

We also import the F1 metric, which is the harmony average of the *precision* and *recall* metrics.

Ranking accuracy. We import two metrics, namely, MRR and NDCG, to evaluate the ranking performance of each model (the recommendation results can be seen as a ranking list of all the candidate items). The position of the first correct item in the recommendation list indicates the mean reciprocal rank (MRR), which is calculated as:

$$MRR = \frac{1}{|U|} \sum_{u \in U} \frac{1}{rank_u^1}, \quad (10)$$

where $rank_u^1$ denotes the position of the first correctly predicted item for user u .

We let rel_i be the relevance of the item at position i for a ranked retrieval, and define the discounted cumulative gain (DCG) at position n as:

$$DCG@n = \sum_{i=1}^n \frac{2^{rel_i} - 1}{\log_2(i + 1)} \quad (11)$$

For unknown items for a target user, we set the relevance to 1 for the items in the test set and to 0 for the others and obtain the ideal DCG score, which is known as the IDCG. Eventually, The NDCG (normalized DCG) score is calculated as the predicted DCG divided by the IDCG.

Hit rate (HR). A recommender should be able to meet people's needs. We believe that a user will be satisfied if they can accept at least one item from the recommendation list, which implies that at least one item was correctly recommended. Hence, we define the overall HR of a model as the proportion of satisfied users in the whole user set, as follows:

$$HR@n = \frac{\# \text{ satisfied users}}{\# \text{ all users}} \quad (12)$$

5.4. Results comparison

Classification performance

Table 4 compares the performance of the models in terms of the precision, recall, and F1 metrics, with a recommendation size of 1, 5, and 10, respectively. The SeqRank model outperforms the other models on all datasets. Specifically, the improved performance of the SeqRank model compared with that of the random walk variant, SeqRank-walk, illustrates the success of the application of the regularized learning framework. Moreover, the SeqRank model outperforms the other sequential models including the variant of SeqRank-norate, which do not consider user ratings, by successfully incorporating users' sequential behavior and rating feedback.

Ranking performance

Fig. 4 displays the trend of the NDCG ranking metric as the recommendation size increases in each dataset. The SeqRank model consistently obtains the best results, thereby indicating that it can obtain a highly precise ranking order for the candidate items. Furthermore,

to emphasize the efficiency of the models in identifying the first truly desired item, we present the MRR metric results in Table 5, which shows that the SeqRank model consistently obtains the best scores on the four datasets.

User satisfaction

To compare overall user satisfaction, we collect and present the HR metric results in Fig. 5. The results show that the SeqRank model has the highest HR scores in all the cases in the datasets except Tmall. The Tmall dataset is with the highest data sparseness and the deep model Caser ranks the based in most cases, but the SeqRank model achieves very closed results and even outperform Caser when recommendation size is very small (like $n < 5$).

We summarize the findings from the above comparisons. First, the SeqRank model and random walk variant iterate in similar ways, but the former performs better than the latter. This outcome demonstrates the success of the implementation of the regularized graph ranking framework. Second, the SeqRank model outperforms the SeqRank-norate (the variant model with no rating input) and other Markov models consistently and deep models in most circumstances, thereby highlighting the effectiveness of combining scaled rating data and sequential recommendations.

5.5. Parameter sensitive

The proposed SeqRank model has two parameters. Parameter α balances the contributions of the graph structure and the user queries, and parameter μ adjusts the contributions of the sequential effects and user preferences. Thus, the proposed model is an integrated model that combines the advantages of the Markov model and CF model. We use the variation in the recall metric to test the sensitivity of the parameters. Each time, we keep one parameter constant and let the other vary between 0.1 and 0.9. The results are presented in Fig. 6.

Overall, the SeqRank model is sensitive to the value of the two parameters. For the parameter α which adjusts the contributions of the graph structure and the input query, we can summarize from Fig. 6(a) to (d) that the optimized parameter value goes higher as the data sparseness increases. According to Eq. (6), the greater the value of α in the iterative, the more the weight given to the graph structure, rather than to the user query. This suggests that in a sparse environment, where initiating precise user queries is challenging, parameter α should be tuned higher for datasets with sparse rating information, such as those from Tmall and Amazon.

For the parameter μ , balancing the contributions of the sequential effect and personalized interests, a higher value indicates more weight given to the sequential effect. From Fig. 6(a) to (d), the optimized value of parameter μ is found with no significant increasing or decreasing trends along with rating sparseness. The reason is that this parameter reflects the "sequential intensity" of user behaviors in a system. The results suggest emphasizing the sequential effect in the MovieLens and Beauty datasets, while prioritizing personalized user interests in the Tmall and MoviesTV datasets. These findings are partly consistent with that of [38], which discovered that MovieLens users have a substantially higher sequential intensity than Tmall users. Interestingly, for the other two Amazon datasets, the Beauty dataset shows much higher "sequential intention" among users (optimized μ value of 0.9) compared to the MoviesTV dataset (optimized μ value of 0.2). This indicates that people have more "sequential intention" when buying beauty products but have more "personal intention" when buying movies or TV products.

6. Conclusion

In this research, we present a sequential e-commerce recommendation model based on a heterogeneous graph ranking method. We point out that though e-commerce sequential buying behavior is typically accompanied by scaled rating feedback, ratings are rarely handled by

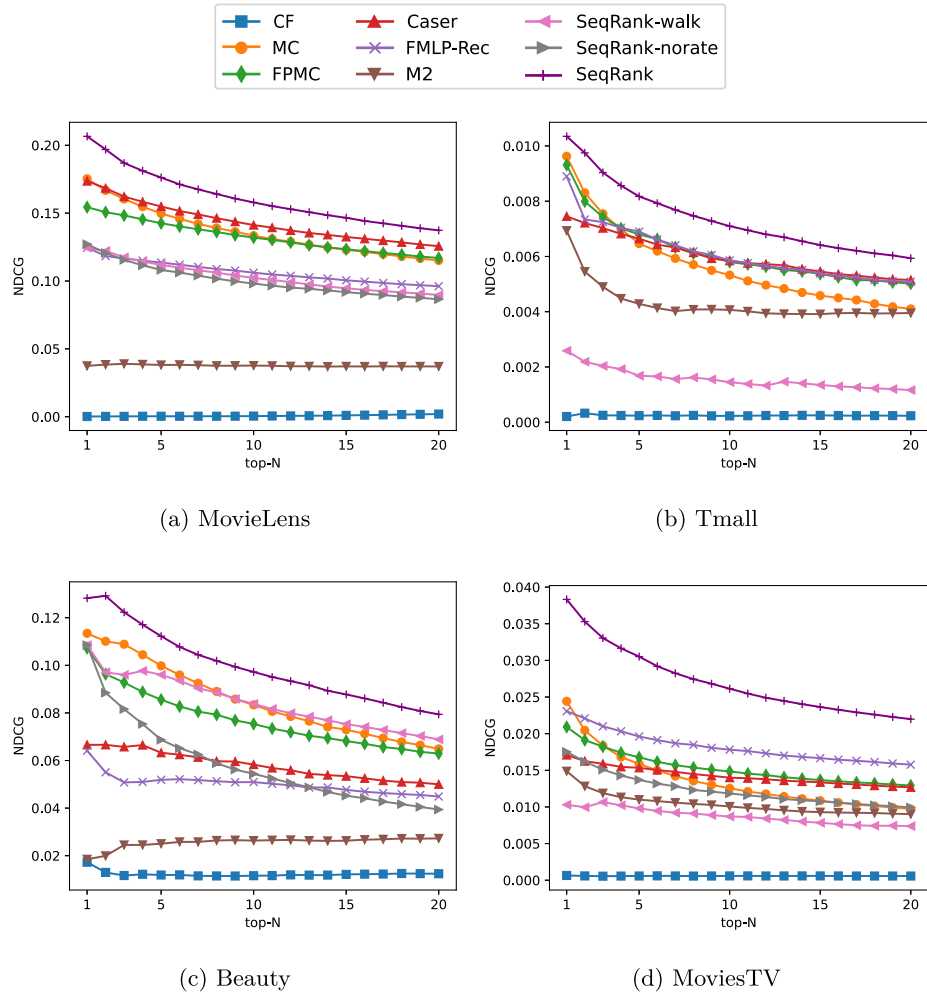


Fig. 4. Ranking performance in terms of NDCG.

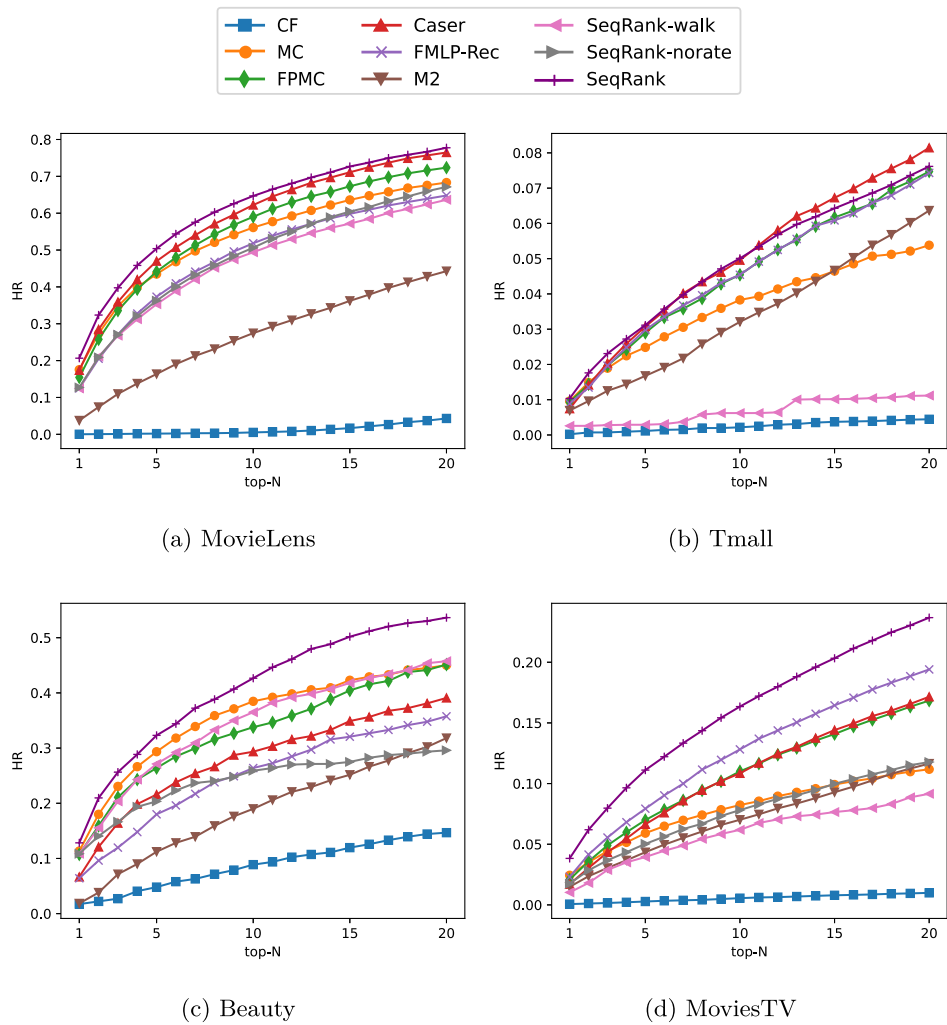


Fig. 5. Results of hit rate (user satisfaction).

Table 4
Classification performance.

		precision@n			recall@n			F1@n		
		n = 1	n = 5	n = 10	n = 1	n = 5	n = 10	n = 1	n = 5	n = 10
Movielens	CF	0.0002	0.0004	0.0006	0.0000	0.0001	0.0002	0.0000	0.0001	0.0003
	MC	<u>0.1753</u>	0.1423	0.1224	<u>0.0152</u>	0.0590	0.0978	<u>0.0280</u>	0.0834	0.1088
	FPMC	0.1544	0.1389	0.1254	0.0135	0.0585	0.1017	0.0248	0.0823	0.1123
	Caser	0.1736	<u>0.1494</u>	<u>0.1324</u>	0.0142	<u>0.0604</u>	<u>0.1059</u>	0.0263	<u>0.0860</u>	<u>0.1176</u>
	FMLP-Rec	0.1245	0.1111	0.1015	0.0102	0.0435	0.0800	0.0189	0.0626	0.0895
	M^2	0.0375	0.0380	0.0375	0.0031	0.0162	0.0300	0.0058	0.0228	0.0333
	SeqRank-Walk	0.1245	0.1079	0.0959	0.0094	0.0361	0.0637	0.0174	0.0541	0.0766
	SeqRank-norate	0.1271	0.1027	0.0909	0.0147	0.0542	0.0938	0.0263	0.0709	0.0923
	SeqRank	0.2066	0.1677	0.1457	0.0173	0.0669	0.1123	0.0319	0.0956	0.1268
	improv.	17.9%	12.2%	10.0%	13.8%	10.8%	6.0%	13.9%	11.2%	7.8%
Tmall	CF	0.0002	0.0002	0.0002	0.0000	0.0003	0.0005	0.0000	0.0003	0.0003
	MC	<u>0.0096</u>	0.0056	0.0044	0.0021	0.0065	0.0099	0.0035	0.0060	0.0061
	FPMC	0.0093	0.0062	0.0051	<u>0.0022</u>	0.0069	0.0111	<u>0.0036</u>	0.0065	0.0070
	Caser	0.0075	0.0064	<u>0.0054</u>	0.0020	<u>0.0078</u>	<u>0.0125</u>	0.0032	<u>0.0070</u>	<u>0.0075</u>
	FMLP-Rec	0.0089	<u>0.0065</u>	0.0052	0.0021	0.0073	0.0114	0.0034	0.0069	0.0072
	M^2	0.0069	0.0037	0.0037	0.0017	0.0042	0.0080	0.0028	0.0040	0.0051
	SeqRank-Walk	0.0026	0.0014	0.0012	0.0006	0.0013	0.0023	0.0009	0.0014	0.0016
	SeqRank	0.0103	0.0075	0.0063	0.0026	0.0086	0.0143	0.0041	0.0080	0.0087
	improv.	7.3%	15.3%	16.7%	18.2%	10.3%	14.4%	13.9%	14.3%	16.0%
Beauty	CF	0.0173	0.0111	0.0111	0.0034	0.0116	0.0253	0.0057	0.0113	0.0154
	MC	0.1134	0.0949	0.0737	<u>0.0223</u>	<u>0.0947</u>	0.1429	0.0373	0.0948	0.0973
	FPMC	0.1073	0.0801	0.0682	0.0210	0.0745	0.1171	0.0352	0.0772	0.0862
	Caser	0.0666	0.0617	0.0552	0.0122	0.0573	0.0979	0.0206	0.0594	0.0706
	FMLP-Rec	0.0641	0.0503	0.0497	0.0113	0.0443	0.0867	0.0192	0.0471	0.0632
	M^2	0.0185	0.0266	0.0276	0.0039	0.0236	0.0534	0.0064	0.0250	0.0364
	SeqRank-Walk	0.1085	0.0942	<u>0.0774</u>	0.0221	0.0932	<u>0.1472</u>	0.0367	0.0937	<u>0.1015</u>
	SeqRank-norate	0.1085	0.0587	0.0432	0.0211	0.0729	0.1045	0.0370	0.0650	0.0611
	SeqRank	0.1282	0.1060	0.0877	0.0274	0.1088	0.1730	0.0452	0.1074	0.1164
	improv.	13.1%	11.7%	13.3%	22.9%	14.9%	17.5%	21.2%	13.3%	14.7%
MoviesTV	CF	0.0006	0.0006	0.0006	0.0001	0.0006	0.0011	0.0002	0.0006	0.0008
	MC	<u>0.0244</u>	0.0138	0.0101	<u>0.0048</u>	0.0124	0.0177	<u>0.0080</u>	0.0130	0.0128
	FPMC	0.0209	0.0157	0.0135	0.0034	0.0124	0.0208	0.0058	0.0139	0.0164
	Caser	0.0171	0.0149	0.0132	0.0024	0.0109	0.0194	0.0043	0.0126	0.0157
	FMLP-Rec	0.0231	<u>0.0186</u>	<u>0.0165</u>	0.0036	<u>0.0153</u>	<u>0.0253</u>	0.0062	<u>0.0161</u>	<u>0.0200</u>
	M^2	0.0149	0.0101	0.0092	0.0021	0.0071	0.0129	0.0037	0.0084	0.0108
	SeqRank-Walk	0.0103	0.0095	0.0081	0.0014	0.0064	0.0110	0.0024	0.0077	0.0093
	SeqRank-norate	0.0175	0.0126	0.0105	0.0036	0.0122	0.0197	0.0060	0.0124	0.0137
	SeqRank	0.0383	0.0285	0.0232	0.0072	0.0253	0.0397	0.0122	0.0268	0.0293
	improv.	56.9%	53.2%	40.6%	50.0%	65.3%	56.9%	52.5%	66.5%	46.5%

Table 5
Ranking performance in terms of MRR.

	MovieLens	Tmall	Beauty	MoviesTV
CF	0.0139	0.0023	0.0430	0.0035
MC	0.3011	0.0200	0.1993	0.0443
FPMC	0.2915	0.0242	0.1889	0.0535
Caser	0.3146	0.0247	0.1451	0.0497
FMLP-Rec	0.2454	0.0237	0.1287	0.0599
M^2	0.1166	0.0189	0.0762	0.0362
SeqRank-walk	0.2424	0.0063	0.1903	0.0301
SeqRank-norate	0.2470	–	0.1594	0.0409
SeqRank	0.3474	0.0263	0.2286	0.0821

existing sequential recommender systems. Our proposed model constructs a unique graph for users, items, and diverse relations, such as user–item undirected rating relation and item–item directed sequential dependence relation, then searches for the nearest item nodes on the heterogeneous graph structure using a query-based iterative ranking algorithm. In addition, the proposed model creates a customized query as input for the target user based on their recent buying history and personalized interests. The proposed model has the advantage of offering a simple but effective way to capture users' time-aware requests and time-invariant preferences, making it useful to e-commerce cross-selling studies. Empirically, we show that the proposed model outperforms Markov models and even deep models at the most cases on four public datasets from different fields, especially in sparse environments. In place of a regularized ranking framework, a deep learning method based on heterogeneous user–item graph data can be developed

in the future. For example, a graph convolutional network [50] can be used for item representation and recommendation.

CRedit authorship contribution statement

Hongyi Mao: Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Data curation. **Mingsong Mao:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Fuhua Mao:** Visualization, Validation, Software, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The work was supported by the National Natural Science Foundation of China (Nos. 72161018, U2268209, 72061005, 61802156), the Educational Commission of Jiangxi Province of China (GJJ200503), Startup Foundation for Distinguished Scholars of Guizhou University of Finance and Economics (2019YJ065).

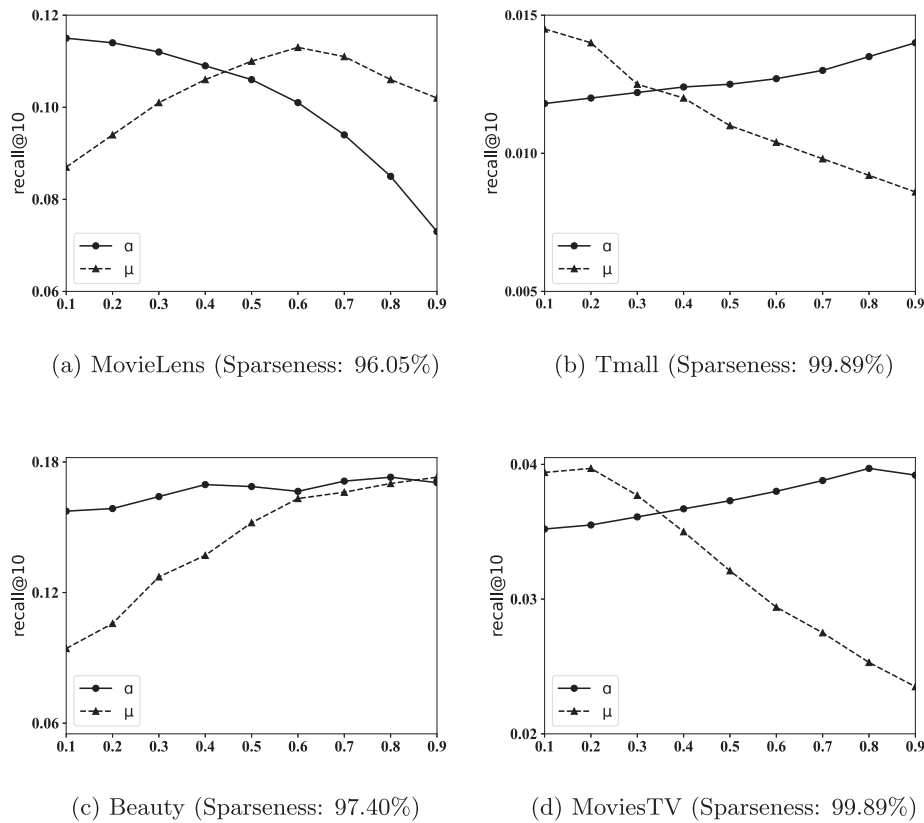


Fig. 6. Parameter sensitivity for different datasets.

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